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**CSC3102 Project 3 - Group P**

**Test Cases:**

**1. Print network list for user-selected student**

Input: Enter choice: 1  
Enter student ID: 1

Output: Student Name: Ehsan Agha Ebrahimi; Student Number: 1  
Student Name: Madelaine Brown; Student Number: 13  
Student Name: Myles Guidry; Student Number: 41  
Student Name: Skylur Fleming; Student Number: 33  
Student Name: Mujtaba Malik; Student Number: 65  
➤ The user successfully printed their selected student's network.

Input: Enter choice: 1  
Enter student ID: -1

Output: Invalid Input.  
➤ The user inputted a negative number that's not a proper student ID and proceeds to input an appropriate ID number.

Input: Enter choice: 1  
Enter student ID: 120941

Output: Invalid Input.  
➤ The user inputted an out-of-range number that's not within the class size and proceeds to input an appropriate ID number.

Input: Enter choice: 1  
Enter student ID: cran

Output: Invalid Input.  
➤ The user inputted a string that's not a proper student ID and proceeds to input an appropriate ID number.

➤ Printing the user-selected student's network tests cases ensures that the program doesn't display error messages when the user inputs negative, non-numeric, or out of range class size student ID values.

**2. Find quickest path from student A to student B**

Input: Enter choice: 2  
Enter start student ID: 1  
Enter end student ID: 100000

Output: Invalid Student IDs.  
➤ The user inputted an out of range class size number and proceeds to input appropriate ID numbers

Input: Enter choice: 2  
Enter start student ID: 100000

Output: Invalid input.  
➤ The user inputted an out of range class size number and proceeds to input an appropriate ID number.

Input: Enter choice: 2  
Enter start student ID: what

Output: Invalid input.  
➤ The user inputted a non-numeric student ID value and proceeds to input an appropriate ID number.

Input: Enter choice: 2  
Enter start student ID: -1  
Enter end student ID: 0

Output: Invalid Student IDs.  
➤ The user inputted improper student ID values and proceeds to input appropriate ID numbers.

Input: Enter choice: 2  
Enter start student ID: 56  
Enter end student ID: 49

Output: Shortest Path: 9 days, 2 nodes  
➤ The user successfully printed the shortest path of the number of days it takes and number of nodes to the path.

➤ Finding the quickest path test cases ensures that the program doesn't exhibit errors when the user inputs out of range class size, non-numeric, and negative students ID values.

**3. Disconnect user-selected student B from A's network**

Input: Enter choice: 2  
Enter student ID: -1  
Enter connection ID to disconnect: -1

Output: Student Could not be found.  
➤ The user inputted invalid student ID values that are not within class size.

Input: Enter choice: 3  
Enter student ID: hi

Output: Invalid input.  
➤ The user inputted a non-numeric student ID and proceeds to input an appropriate ID value.

Input: Enter choice: 3  
Enter student ID: 114  
Enter connection ID to disconnect: 102

Output: 1. Print network list for student  
2. Find quickest path between students  
3. Disconnect students  
4. Increase wait days for student  
5. Decrease wait days for student  
6. Exit  
➤ The user successfully removed an edge of student B from student A's network.

- Disconnecting an edge of user-selected student B from student A's network test cases ensures that the program doesn't present errors to the user when inputting out-of-range class size, negative, and non-numeric ID values. Changes to the graph are adjusted accordingly to display the network of the user-selected student and quickest path of user-selected students.

**4. Increase wait days of user-selected student**

Input: Enter choice: 4  
Enter student ID: -300  
Enter days to increase: 50

Output: Student not found.  
➤ The user inputted an invalid student ID that is not within class size.

Input: Enter choice: 4  
Enter student ID: tree

Output: Invalid input  
➤ The user inputted a non-numeric student ID value and proceeds to input an appropriate ID value.

Input: Enter choice: 4  
Enter student ID: 300  
Enter days to increase: gam

Output: Invalid input  
➤ The user inputted a non-numeric day value and proceeds to input an appropriate ID value.

Input: Enter choice: 4  
Enter student ID: 10  
Enter days to increase: 30

Output: Wait days updated to 42  
➤ The user successfully added wait days for user-selected student.

- Increasing number of wait days for user-selected student ensures that the program doesn't show errors to the user when they input non-numeric and out-of-range class ID and day values. Changes to the graph are adjusted accordingly to display the network of the user-selected student and quickest path of user-selected students.

**5. Decrease wait days of user-selected student**

Input: Enter choice: 5  
Enter student ID: air

Output: Invalid input.  
➤ The user inputted a non-numeric student ID value and proceeds for an appropriate ID value.

Input: Enter choice: 5  
Enter student ID: -2000  
Enter days to decrease: -43

Output: Invalid input.  
➤ The user inputted an invalid student ID that is not within class size and invalid day value to decrease.

Input: Enter choice: 5  
Enter student ID: 25  
Enter days to decrease: rock

Output: Invalid input.  
➤ The user inputted an invalid non-numeric day value to decrease.

Input: Enter choice: 5  
Enter student ID: 86  
Enter days to decrease: 10

Output: Wait days updated to 7  
➤ The user successfully decreased the number of wait days for user-selected student.

- Decreasing the number of wait days for the user-selected student ensures that the program doesn't display errors to the user when inputting non-numeric and out-of-range class size ID and day values. Changes to the graph are adjusted accordingly to display the network of the user-selected student and quickest path of user-selected students.

**Execution:**

Input: 1. Print network list for student  
2. Find quickest path between students  
3. Disconnect students  
4. Increase wait days for student  
5. Decrease wait days for student  
6. Exit  
Enter choice: -75

Output: Invalid choice.  
1. Print network list for student  
2. Find quickest path between students  
3. Disconnect students  
4. Increase wait days for student  
5. Decrease wait days for student  
6. Exit

Enter choice:  
➤ The user inputted an invalid choice operation and proceeds for an appropriate choice operation.